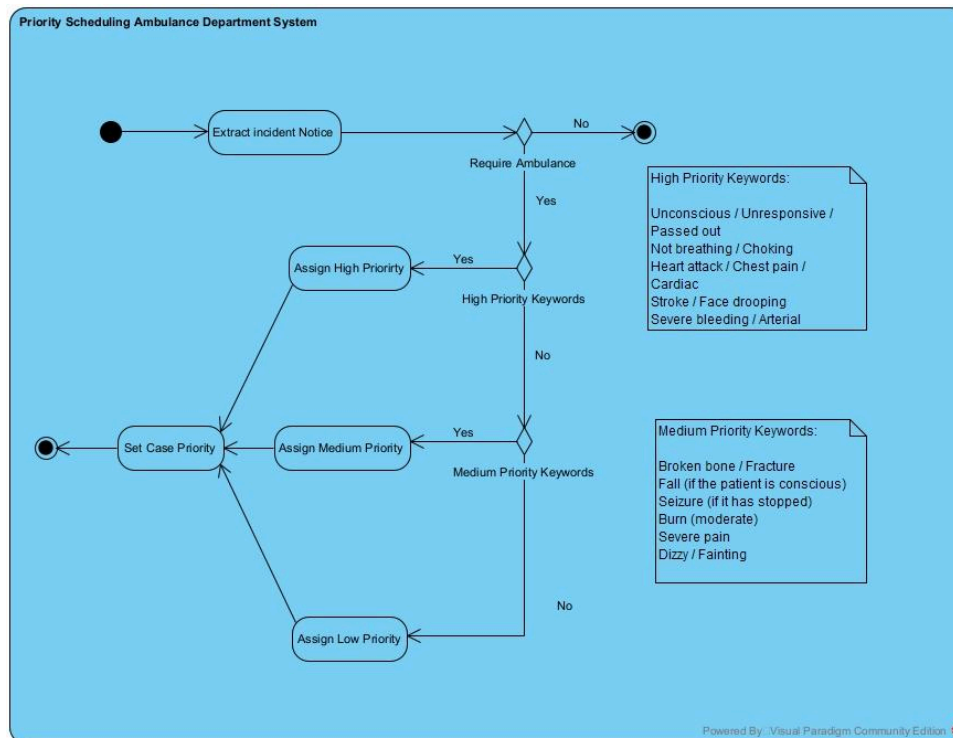
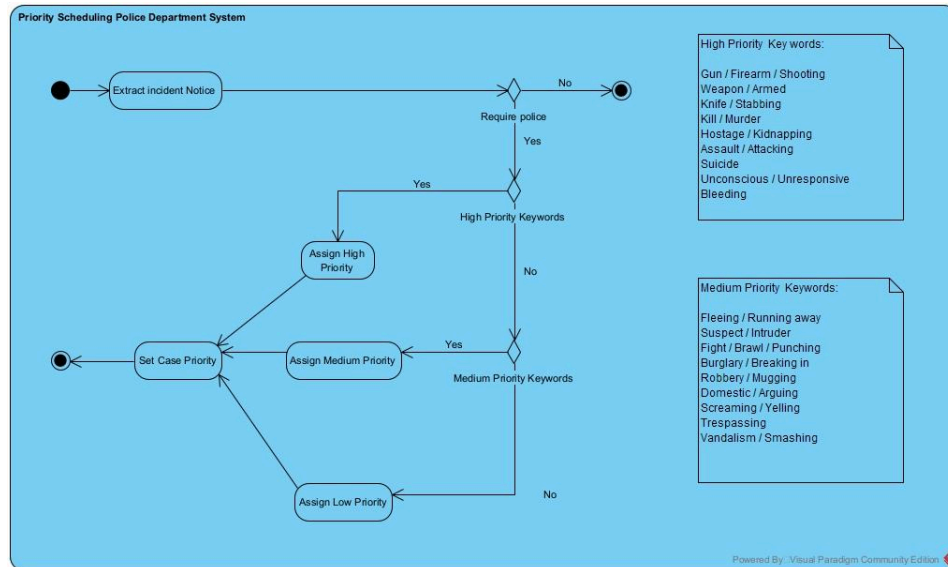
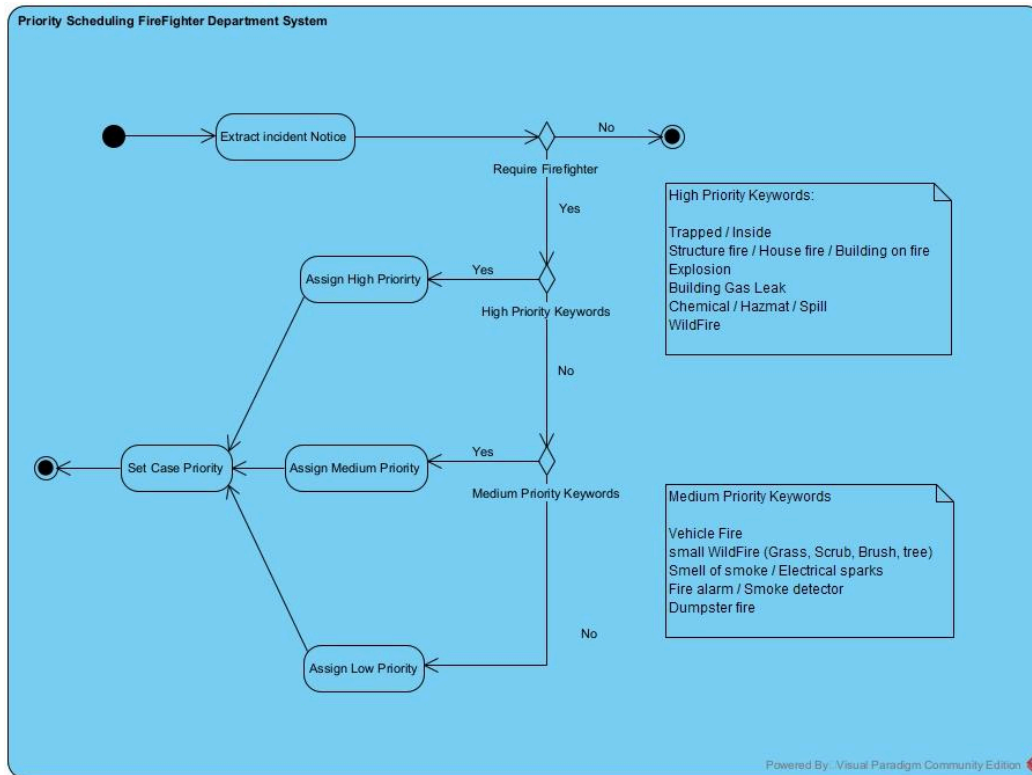


# Appendix

## Appendix 1. Priority/severity activity diagrams for Medical, Police, and Fire





## Appendix 2. Critical and Non-Critical modules

| Module                     | Critical / Non Critical |
|----------------------------|-------------------------|
| Record Log & Case Creation | Critical                |
| Unit & Resource Management | Critical                |
| Priority & Routing Units   | Critical                |
| Case Status Of Units       | Critical                |
| Department DashBoard       | Non-Critical            |
| Notification Service       | Non-Critical            |
| Search Logs                | Non-Critical            |

### Appendix 3. Roles and role permissions

Any feature that is not listed in the Interface Access, and Permission / Actionables columns corresponding to each user role is denied access.

| User role                                       | Interface Access                                                                                                                                                                                                                                                                               | Permissions / Actionables                                                                                                                                                                                                            |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dispatcher                                      | <ul style="list-style-type: none"><li>• Access to Dashboard view of all active case (Case ID, location, Priority, required response, assigned unit, status) and check view recently created cases</li><li>• Access to Record Emergency page</li><li>• Access to the history log page</li></ul> | <ul style="list-style-type: none"><li>• Record details (From the caller)</li><li>• Create emergency case</li><li>• Search any logs from the history log page</li></ul>                                                               |
| Emergency Department<br>(Police, Fire, Medical) | <ul style="list-style-type: none"><li>• Access to Dashboard (Display the assigned case related to that specific department)</li><li>• View response unit details for that specific department in the Dashboard and Units page.</li><li>• View notifications (In Dashboard)</li></ul>           | <ul style="list-style-type: none"><li>• Update unit details such as location and unit count.</li><li>• Unit availability is updated automatically by the system.</li></ul>                                                           |
| Unit (Police, Fire, Medical)                    | <ul style="list-style-type: none"><li>• View the assignment interface showing the case assigned to that specific response unit and its incident details.</li></ul>                                                                                                                             | <ul style="list-style-type: none"><li>• Sign off after it completes its response.</li><li>• Signing off releases that unit.</li><li>• The case is automatically closed once all units assigned to that case has signed off</li></ul> |

## Appendix 4. Core tasks for each user role

| User Role                                    | Core Tasks                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dispatcher                                   | <ul style="list-style-type: none"><li>• Case creation flow (Receive call, then allowed to enter all required fields details specified from the caller, select the unit type, and create the case.)</li><li>• Search case (In the history log interface, search log by call name, case ID or Record date, to view cases.)</li></ul>                                                                          |
| Emergency Department (Police, Fire, Medical) | <ul style="list-style-type: none"><li>• Unit Management (In the Units interface, view all response units belonging to that department and update editable details such as location and personnel count.)</li><li>• Monitor Dashboard (In the Dashboard interface, view cases requiring that department, response unit availability, assigned units, waiting cases, and assignment notifications.)</li></ul> |
| Unit (Police, Fire, Medical)                 | <ul style="list-style-type: none"><li>• Sign-off (in the assignment interface, when the response unit has completed its response, it can sign off from the case)</li><li>• View case (view the incident details and assignment details for the case currently assigned to that specific response unit)</li></ul>                                                                                            |

## Appendix 5. Assessment 1 Functional and Non-Functional Requirements

### Functional requirement from assignment one

- **FR-01** The system shall provide an interface for the dispatcher to record an emergency call, requiring the caller information, incident type, incident location, incident description, timestamp, and relevant dispatch units required to go on the scene.
- **FR-02** The system shall allow the dispatcher to create an emergency case from a recorded call, capturing all the details provided in FR-01.
- **FR-03** The system shall allow each emergency response department to view an interface and update the availability status, unit type, location, identifier, and count of the response units managed by that department.
- **FR-04** The system shall automatically determine the priority level of a newly created emergency case using the predefined decision process for the relevant emergency response department, based on the incident information recorded by the dispatcher. The priority determination processes for Medical, Police, and Fire cases are defined in Appendix 1.
- **FR-05** The system shall automatically route a case to the emergency response departments based on which dispatch units were assigned to the case.
- **FR-06** Upon a case being routed to a department, the department's system shall automatically assign the units from that specific department based on whether the unit is available. If multiple units are available, the system shall assign the first unit in the department's unit list. If no units are available, the case shall remain in a queue until a unit becomes available.
- **FR-07** The system shall provide each emergency response department with a dashboard displaying the current cases assigned to that department. Any changes to the case information should be updated on the dashboard without needing the user to manually refresh.
- **FR-08** The system shall update the status of each case by following these states: "Open" when a case is created, not yet assigned to a unit, "In Progress" when at least one response unit is assigned and has not signed off, and "Closed" when all response units assigned to the case have signed off. If all assigned units are removed before completing the response, the case shall return to "Open".

- **FR-09** When a unit is assigned to an active case, the system shall change the unit's availability status to "not available". When that response unit signs off from the case after completing its response, the system shall change its status to "available".
- **FR-10** The system shall notify the relevant unit by a notification displayed on the dashboard of the system. The notification should contain the assigned case with its case details upon assignment.
- **FR-11** The system shall allow each response unit assigned to a case to sign off when it has completed its response. The system shall automatically close the case once all response units assigned to that case have signed off.
- **FR-12** The system shall allow the dispatcher to search the history of logs using any combination of date, caller name, or case ID, returning all logs that match at least one entered field.

#### **Non-functional Requirements from assignment one**

- **NFR-P01:** When a completed case is submitted for routing, the system should send the case to all relevant emergency response departments within 2 seconds.
- **NFR-P02:** The system shall support at least 100 concurrent users while maintaining a response time of 2 seconds or less when navigating between interfaces or refreshing a view.
- **NFR-M01:** The system shall be built using a modular architecture such that the core dispatch functions (case creation, unit assignment, status updates) are not affected by non-critical modules (notification, search logs). **Appendix 2**
- **NFR-M02:** The system shall be designed in a way that allows new user roles and departments to be added without requiring modification of existing classes.
- **NFR-R01:** Data from a case shall not be lost when it is being sent from the dispatcher to the selected emergency department.
- **NFR-R02:** The system should maintain at least 99.9% availability over a week; this includes the systems for Caller, Dispatcher, Field Responders, and Emergency Departments. Planned maintenance work does not count towards this downtime for any area.
- **NFR-R03:** All case log entries and incident reports submitted through the system shall be saved without losing any of the data entered

- **NFR-R04:** When 100 cases are open simultaneously by the dispatchers, response units and emergency departments, the system shall continue accepting, displaying, and updating cases without crashing.
- **NFR-S01:** Any data that passes between users and the system must be encrypted to prevent unauthorized access.
- **NFR-S02:** The system shall restrict each authenticated user to only the system features and information permitted for their assigned role. The role permissions are defined in **Appendix 3**.
- **NFR-U01:** The system shall display a tailored interface of the system for each authenticated user's assigned role, displaying only the tools and information required for that role. The roles and role permissions are shown in **Appendix 3**
- **NFR-U02:** The system shall return all error messages in plain language so they can be easily understood. It should also provide an action that can resolve the error and let the user know what to do. Technical error messages such as exceptions, stack traces, and debug information should never be displayed to the users.
- **NFR-U03:** After no more than 1 hour of training, each user shall be able to complete all defined core tasks for their assigned role without assistance. The core tasks for each role are listed in **Appendix 4**.

## Appendix 6. Assessment 2 Functional and Non-Functional Requirements

### Functional requirement from assignment two

- **FR-01:** The system shall provide a dispatcher interface for recording caller details, incident type, description, location, coordinates, reported severity and required response departments. The system shall record the submission timestamp.
- **FR-02** When valid emergency information is submitted, the system shall create one case containing the recorded details, a unique case identifier and a generated timestamp.
- **FR-03** The system shall allow each emergency response department to view the response units belonging to that department and update permitted unit information, including location, coordinates, and personnel count. Unit availability shall be controlled automatically by the system based on assignment and sign-off.
- **FR-04** The system shall automatically determine the priority level of a newly created emergency case using the predefined decision process for the relevant emergency response department, based on the incident information recorded by the dispatcher. The priority determination processes for Medical, Police, and Fire cases are defined in Appendix 1. If no keywords are matching, then the reported severity will be used. For multi-department cases, the highest result shall be used. An authorised dispatcher may override the priority after providing a reason.
- **FR-05** The system shall route a case to every department selected by the dispatcher. The case shall be visible to the selected departments and hidden from unrelated departments.
- **FR-06** For each required department, the system shall assign the nearest compatible available unit using coordinates and distance. Unavailable units shall be excluded. If no compatible unit is available, the case shall remain waiting until one becomes available.
- **FR-07** The system shall provide each department with a dashboard containing only cases relevant to that department. The dashboard shall reload every five seconds so new information appears without having to manually refresh.
- **FR-08** A case shall be Open while waiting with no active assignment, In Progress while at least one required response is active, and Closed after every required response has signed off.

- **FR-09** When a unit is assigned to an active case, the system shall change the unit's availability status to "not available". When the response unit signs off from the case, the system shall change its status to "available" and may assign it to the oldest compatible waiting case.
- **FR-10** When a unit is assigned, the system shall display a notification containing the assigned case number, incident type and location. A notification failure shall not reverse successful case creation or case assignment.
- **FR-11** The system shall allow each response unit assigned to a case to sign off when it has completed its response. The system shall automatically close the case once all response units assigned to that case have signed off.
- **FR-12** The system shall allow the dispatcher to search the history of logs using any combination of date, caller name, or case ID, returning all logs that match at least one entered field.
- **FR-13** The system shall persist audit events for important actions, including case creation, routing selections, assignment, unit updates, priority overrides, sign-off and closure. Each event shall include the action, actor, timestamp and relevant case or unit details.

#### **Non-functional Requirements from assignment two**

- **NFR-P01:** A locally submitted case shall be stored, routed and assigned to all relevant departments within two seconds under the prototype test conditions.
- **NFR-P02:** The system shall support at least 100 concurrent users while maintaining a response time of two seconds or less when navigating or refreshing an interface.
- **NFR-M01:** The system shall be built using a modular architecture such that the core dispatch functions (case creation, unit assignment, status updates) are not affected by non-critical modules (notification, search logs). A failure in the notification component shall not undo successful case storage or unit assignment. **Appendix 2.**
- **NFR-M02:** New roles and departments should be introducible without changing the existing core rules for case creation, routing, assignment and lifecycle status.
- **NFR-R01:** Case information made visible to each selected department shall remain complete and consistent with the case submitted by the dispatcher.
- **NFR-R02:** The deployed system should maintain at least 99.9% availability over a seven-day period, excluding planned maintenance.

- **NFR-R03:** Cases, units, assignments, notifications, accounts and audit events shall remain retrievable without missing or changed fields after the application and database are restarted.
- **NFR-R04:** When 100 cases are open simultaneously by the dispatchers, response units and emergency departments, the system shall continue accepting, displaying, and updating cases without crashing.
- **NFR-S01:** Selected sensitive account, caller, incident and location values stored in SQLite shall be encrypted using AES-256-GCM. A production deployment shall use TLS 1.2 or higher for information transmitted between users and the server.
- **NFR-S02:** The system shall restrict each authenticated user to only the pages, actions and information permitted for their assigned role and department or unit scope. Direct requests to unauthorised pages or records shall be denied. The role permissions are defined in **Appendix 3**.
- **NFR-S03:** Account passwords shall be stored as salted PBKDF2 hashes rather than plaintext. Correct passwords shall be accepted, and incorrect passwords shall be rejected.
- **NFR-U01:** The system shall display a tailored interface of the system for each authenticated user's assigned role, displaying only the tools and information required for that role. The roles and role permissions are shown in **Appendix 3**
- **NFR-U02:** User-facing validation and error messages shall use understandable language, identify the problem where appropriate, and shall not expose exceptions, stack traces or debugging information.
- **NFR-U03:** After no more than 1 hour of training, each user shall be able to complete all defined core tasks for their assigned role without assistance. The core tasks for each role are listed in **Appendix 4**.

## Appendix 7. Assessment 2, 17 selected test cases in detail

### TC-01: Create a case with valid information

|                   |                                                                                                                                                                  |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Requirements      | FR-01, FR-02                                                                                                                                                     |
| Test level / type | Integration                                                                                                                                                      |
| Test condition    | Positive                                                                                                                                                         |
| Risk priority     | High                                                                                                                                                             |
| Preconditions     | <ol style="list-style-type: none"><li>1. The prototype is running.</li><li>2. The dispatcher has authorisation on the information recording interface.</li></ol> |
| Test data         | John Smith, vehicle collision, High severity, and departments (Police and Medical).                                                                              |
| Steps             | <ol style="list-style-type: none"><li>1. Submit the case.</li><li>2. Retrieve the case.</li><li>3. Check its details and timestamp.</li></ol>                    |
| Expected result   | One case is saved with the correct details, case number and timestamp.                                                                                           |
| Observed result   | A case was saved successfully with the expected details.                                                                                                         |
| Execution status  | Pass                                                                                                                                                             |
| Trace to C#       | Test/PrototypeTesting.cs<br>PrototypeTesting.TC01_CompleteEmergencyDetails_CreateOneCaseWithRecordedData                                                         |

### TC-02: Reject incomplete case information

|                   |                                                                                     |
|-------------------|-------------------------------------------------------------------------------------|
| Requirements      | FR-01                                                                               |
| Test level / type | Integration                                                                         |
| Test condition    | Invalid Input                                                                       |
| Risk priority     | High                                                                                |
| Preconditions     | <ol style="list-style-type: none"><li>1. Empty in-memory case repository.</li></ol> |
| Test data         | A missing location info.                                                            |

|                  |                                                                                                                    |
|------------------|--------------------------------------------------------------------------------------------------------------------|
| Steps            | 1. Submit a case without a location.<br>2. Submit a case without a department.<br>3. Check errors and saved cases. |
| Expected result  | Both submissions are rejected, and no case is saved.                                                               |
| Observed result  | Both inputs were rejected and no case was saved.                                                                   |
| Execution status | Pass                                                                                                               |
| Trace to C#      | Test/PrototypeTesting.cs<br>PrototypeTesting.TCO2_IncompleteEmergencyDetails_AreRejectedWithout<br>CreatingCase    |

#### TC-03: Create and assign a multi-department case

|                   |                                                                                                                                 |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Requirements      | FR-02, FR-04, FR-05, FR-06, FR-10                                                                                               |
| Test level / type | Integration                                                                                                                     |
| Test condition    | Positive                                                                                                                        |
| Risk priority     | Critical                                                                                                                        |
| Preconditions     | 1. In-memory departments.<br>2. units are ready.                                                                                |
| Test data         | Collision case, High severity, and Police and Medical.                                                                          |
| Steps             | 1. Submit the case.<br>2. Check priority, status, assignments and notifications.                                                |
| Expected result   | The case should have High priority, status show “In Progress” as well as POL-01 and MED-01 assigned and notifications recorded. |
| Observed result   | The expected priority, assignments and notifications were recorded.                                                             |
| Execution status  | Pass                                                                                                                            |
| Trace to C#       | Test/PrototypeTesting.cs<br>PrototypeTesting.TCO3_Submission_CreatesPrioritisesStoresAndAssignsMulti<br>DepartmentCase          |

#### TC-04: Calculate priority from keywords and severity

|              |       |
|--------------|-------|
| Requirements | FR-04 |
|--------------|-------|

|                   |                                                                                                                                                             |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test level / type | Unit                                                                                                                                                        |
| Test condition    | Positive                                                                                                                                                    |
| Risk priority     | Critical                                                                                                                                                    |
| Preconditions     | 1. A priority calculation component is available.                                                                                                           |
| Test data         | 85 keyword and severity examples for Police, Medical and Fire.                                                                                              |
| Steps             | 1. Create a case for each data row.<br>2. Calculate priority.<br>3. Compare with expected priority.                                                         |
| Expected result   | All 85 cases produce their expected priorities.                                                                                                             |
| Observed result   | All 85 cases successfully returned the expected priorities.                                                                                                 |
| Execution status  | Pass                                                                                                                                                        |
| Trace to C#       | Test/PrototypeTesting.cs<br>PrototypeTesting.TCO4_AppendixDepartmentKeywordsTakePrecedence_The<br>nSeverityIsFallback; data provider: AppendixPriorityCases |

#### TC-05: Complete in-memory dispatch within two seconds

|                   |                                                                                          |
|-------------------|------------------------------------------------------------------------------------------|
| Requirements      | NFR-PO1                                                                                  |
| Test level / type | Performance                                                                              |
| Test condition    | Positive                                                                                 |
| Risk priority     | Critical                                                                                 |
| Preconditions     | 1. In-memory dispatch service is ready.                                                  |
| Test data         | Incident type Building fire and Department Fire, Medical and Police.                     |
| Steps             | 1. Start the timer.<br>2. Dispatch the case.<br>3. Stop the timer and check assignments. |
| Expected result   | Dispatch finishes in under two seconds with the required units assigned.                 |

|                  |                                                                                                                               |
|------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Observed result  | The dispatch system successfully completes within 2 seconds, and the required units are assigned.                             |
| Execution status | Pass                                                                                                                          |
| Trace to C#      | Test/PrototypeTesting.cs<br>PrototypeTesting.TC05_InProcessMultiDepartmentSubmission_CompletesWithinTwoSecondsWithoutDataLoss |

#### TC-06: Skip an unavailable response unit

|                   |                                                                                                                 |
|-------------------|-----------------------------------------------------------------------------------------------------------------|
| Requirements      | FR-06, FR-09                                                                                                    |
| Test level / type | Integration                                                                                                     |
| Test condition    | Negative                                                                                                        |
| Risk priority     | Critical                                                                                                        |
| Preconditions     | 1. Closest Police unit P-01 is already occupied.                                                                |
| Test data         | Units P-01 unavailable, P-02 and P-03 available and P-03 is nearer.                                             |
| Steps             | 1. Submit a Police case.<br>2. Check which unit is assigned.<br>3. Check availability.                          |
| Expected result   | P-03 is assigned; P-01 stays unavailable and P-02 stays available.                                              |
| Observed result   | The nearest available unit, P-03, was assigned.                                                                 |
| Execution status  | Pass                                                                                                            |
| Trace to C#       | Test/PrototypeTesting.cs<br>PrototypeTesting.TC06_Assignment_SkipsUnavailableUnitAndSelectsClosestAvailableUnit |

#### TC-07: Queue a case until a Fire unit is released

|                   |                     |
|-------------------|---------------------|
| Requirements      | FR-06, FR-08, FR-09 |
| Test level / type | Integration         |
| Test condition    | Negative            |
| Risk priority     | Critical            |

|                  |                                                                                                                                  |
|------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Preconditions    | 1. Two Fire units are occupied.                                                                                                  |
| Test data        | Fire units F-01 and F-02 and a third Fire case.                                                                                  |
| Steps            | 1. Submit the third case.<br>2. Check that it is waiting.<br>3. Sign off F-01 from its case.<br>4. Check the waiting case again. |
| Expected result  | The case is Open with no unit, then becomes In Progress with F-01 assigned.                                                      |
| Observed result  | The waiting case received the F-01 unit after signing off.                                                                       |
| Execution status | Pass                                                                                                                             |
| Trace to C#      | Test/PrototypeTesting.cs<br>PrototypeTesting.TC07_NoAvailableUnit_QueueCaseUntilAUnitSignsOff                                    |

#### TC-08: Prevent duplicate assignments during concurrent submissions

|                   |                                                                                                       |
|-------------------|-------------------------------------------------------------------------------------------------------|
| Requirements      | FR-09                                                                                                 |
| Test level / type | Integration                                                                                           |
| Test condition    | Negative                                                                                              |
| Risk priority     | Critical                                                                                              |
| Preconditions     | 1. One Medical unit is available.                                                                     |
| Test data         | Two concurrent Medical requests and 25 repeated attempts.                                             |
| Steps             | 1. Submit both requests at the same time.<br>2. Check case and unit states.<br>3. Repeat 25 times.    |
| Expected result   | Only one case gets the unit, the other remains Open.                                                  |
| Observed result   | No duplicate unit assignments occurred in the test.                                                   |
| Execution status  | Pass                                                                                                  |
| Trace to C#       | Test/PrototypeTesting.cs<br>PrototypeTesting.TC08_ConcurrentSubmissions_CannotAssignOneUnitToTwoCases |

#### TC-09 Show only relevant cases in a department dashboard

|                   |                                                                                                                           |
|-------------------|---------------------------------------------------------------------------------------------------------------------------|
| Requirements      | FR-05, FR-07                                                                                                              |
| Test level / type | Integration                                                                                                               |
| Test condition    | Positive                                                                                                                  |
| Risk priority     | High                                                                                                                      |
| Preconditions     | 1. In-memory dashboard model is available.                                                                                |
| Test data         | A Police case, a Fire case, then another Police case.                                                                     |
| Steps             | 1. Load the Police dashboard.<br>2. Create another Police case.<br>3. Reload and inspect the cases.                       |
| Expected result   | Only Police cases appear, including the new Police case.                                                                  |
| Observed result   | The Police cases appeared and the Fire case was excluded.                                                                 |
| Execution status  | Pass                                                                                                                      |
| Trace to C#       | Test/PrototypeTesting.cs<br>PrototypeTesting.TC09_DepartmentDashboard_ContainsOnlyItsCasesAndIncludesNewCasesAfterRefresh |

#### **TC-10: Close a case only after every unit signs off**

|                   |                                                                                                             |
|-------------------|-------------------------------------------------------------------------------------------------------------|
| Requirements      | FR-08, FR-11                                                                                                |
| Test level / type | Integration                                                                                                 |
| Test condition    | Positive                                                                                                    |
| Risk priority     | Critical                                                                                                    |
| Preconditions     | 1. A case has Police and Medical units assigned.                                                            |
| Test data         | Sign off Police first, then Medical.                                                                        |
| Steps             | 1. Sign off Police.<br>2. Check the case status.<br>3. Sign off Medical.<br>4. Check the case status again. |

|                  |                                                                                               |
|------------------|-----------------------------------------------------------------------------------------------|
| Expected result  | The case remains In Progress after the first sign-off and closes after the second.            |
| Observed result  | The case closed only after both units signed off.                                             |
| Execution status | Pass                                                                                          |
| Trace to C#      | Test/PrototypeTesting.cs<br>PrototypeTesting.TC10_CaseRemainsInProgressUntilFinalUnitSignsOff |

#### TC-11: Search emergency case history

|                   |                                                                                                         |
|-------------------|---------------------------------------------------------------------------------------------------------|
| Requirements      | FR-12                                                                                                   |
| Test level / type | Unit                                                                                                    |
| Test condition    | Positive                                                                                                |
| Risk priority     | Critical                                                                                                |
| Preconditions     | 1. Four cases are stored in memory.                                                                     |
| Test data         | Search for Alex, Jordan's case number and 22 Aug 2026.                                                  |
| Steps             | 1. Search using the three criteria.<br>2. Check matching and excluded cases.                            |
| Expected result   | Three matching cases appear, the fourth case would not appear in the log.                               |
| Observed result   | The three expected cases were returned.                                                                 |
| Execution status  | Pass                                                                                                    |
| Trace to C#       | Test/PrototypeTesting.cs<br>PrototypeTesting.TC11_HistorySearch_ReturnsMatchesForCallerCaseIdOr<br>Date |

#### TC-13: Reject a priority override without a reason

|                   |                                           |
|-------------------|-------------------------------------------|
| Requirements      | FR-04, FR-13                              |
| Test level / type | Integration                               |
| Test condition    | Invalid Input                             |
| Risk priority     | High                                      |
| Preconditions     | 1. An existing Low-priority Medical case. |

|                  |                                                                                                                              |
|------------------|------------------------------------------------------------------------------------------------------------------------------|
| Test data        | Override to High with a reason of three spaces.                                                                              |
| Steps            | 1. Attempt the override.<br>2. Check the priority and audit events.                                                          |
| Expected result  | The override is rejected, the priority stays Low and no override audit event is added.                                       |
| Observed result  | The override was rejected without changing priority or audit records.                                                        |
| Execution status | Pass                                                                                                                         |
| Trace to C#      | Test/PriorityOverrideTests.cs<br>PriorityOverrideTests.MissingReason_IsRejectedWithoutChangingPriorityOrWritingOverrideAudit |

#### TC-14: Keep audit records after reopening the database

|                   |                                                                                                                                                        |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Requirements      | FR-13                                                                                                                                                  |
| Test level / type | Integration                                                                                                                                            |
| Test condition    | Positive                                                                                                                                               |
| Risk priority     | High                                                                                                                                                   |
| Preconditions     | 1. Temporary SQLite database<br>2. Fire unit available                                                                                                 |
| Test data         | Include Fire case, actors dispatch02 and fir01.                                                                                                        |
| Steps             | 1. Create and close the case.<br>2. Reopen SQLite.<br>3. Read the audit records.                                                                       |
| Expected result   | Four records remain in the database, case close should be first and case created should be last.                                                       |
| Observed result   | The four expected audit records were stored and available when reopening as well as in the correct order based on the timestamp from newest to oldest. |
| Execution status  | Pass                                                                                                                                                   |

|             |                                                                                                       |
|-------------|-------------------------------------------------------------------------------------------------------|
| Trace to C# | Test/AuditLoggingTests.cs<br>AuditLoggingTests.AuditEvents_SurviveDatabaseRestartAndRemainNewestFirst |
|-------------|-------------------------------------------------------------------------------------------------------|

#### TC-15: Keep case and assignment data after reopening the database

|                   |                                                                                                             |
|-------------------|-------------------------------------------------------------------------------------------------------------|
| Requirements      | NFR-RO3                                                                                                     |
| Test level / type | Integration                                                                                                 |
| Test condition    | Positive                                                                                                    |
| Risk priority     | Critical                                                                                                    |
| Preconditions     | 1. The temporary SQLite database is ready.                                                                  |
| Test data         | Caller Name: Alex Morgan, Incident type: vehicle collision, severity: High and response department: Police. |
| Steps             | 1. Create and assign the case.<br>2. Reopen SQLite.<br>3. Retrieve the case and unit.                       |
| Expected result   | Case details, High priority, In Progress status and unavailable assigned units are preserved.               |
| Observed result   | The expected case and unit data were restored.                                                              |
| Execution status  | Pass                                                                                                        |
| Trace to C#       | Test/SqlitePersistenceTests.cs<br>SqlitePersistenceTests.CreatedCaseAndAssignment_SurviveDatabaseRestart    |

#### TC-16: Check password hashing and verification

|                   |                                                                         |
|-------------------|-------------------------------------------------------------------------|
| Requirements      | NFR-SO3                                                                 |
| Test level / type | Unit                                                                    |
| Test condition    | Positive & Negative                                                     |
| Risk priority     | High                                                                    |
| Preconditions     | 1. PasswordHasher is available.                                         |
| Test data         | Correct password: test-password, second wrong password: wrong-password. |

|                  |                                                                                                               |
|------------------|---------------------------------------------------------------------------------------------------------------|
| Steps            | 1. Hash the correct password twice.<br>2. Compare salts and hashes.<br>3. Verify correct and wrong passwords. |
| Expected result  | Salts and hashes differ when the correct password succeeds and wrong password fails.                          |
| Observed result  | Different salts are generated and the correct hash passed and the incorrect hash failed.                      |
| Execution status | Pass                                                                                                          |
| Trace to C#      | Test/SecurityTests.cs<br>SecurityTests.PasswordHasher_UsesDifferentSaltAndVerifiesOnlyTheCorrectPassword      |

#### TC-17: Reject out-of-range coordinates

|                   |                                                                                                          |
|-------------------|----------------------------------------------------------------------------------------------------------|
| Requirements      | FR-01                                                                                                    |
| Test level / type | Unit                                                                                                     |
| Test condition    | Boundary & Invalid Input                                                                                 |
| Risk priority     | High                                                                                                     |
| Preconditions     | 1. Case and Unit objects can be created directly.                                                        |
| Test data         | Case latitude set to 91 and Unit longitude set to 181.                                                   |
| Steps             | 1. Create a Case with latitude 91.<br>2. Create a Unit with longitude 181.<br>3. Check both exceptions.  |
| Expected result   | Both inputs throw ArgumentOutOfRangeException with the expected parameter names.                         |
| Observed result   | Both invalid coordinates produced the expected exceptions.                                               |
| Execution status  | Pass                                                                                                     |
| Trace to C#       | Test/DistanceAssignmentTests.cs<br>DistanceAssignmentTests.InvalidCoordinates_AreRejectedByDomainObjects |

#### TC-18: Continue dispatch when notification fails

|                  |                                                                                                                  |
|------------------|------------------------------------------------------------------------------------------------------------------|
| Requirements     | FR-10                                                                                                            |
| Test level/type  | Integration                                                                                                      |
| Test condition   | Error Handling                                                                                                   |
| Risk priority    | Critical                                                                                                         |
| Preconditions    | 1. In-memory dispatch service uses a notifier that throws.                                                       |
| Test data        | Medical case and notification service unavailable.                                                               |
| Steps            | 1. Submit the case.<br>2. Let the notifier throw.<br>3. Check the saved case and unit.                           |
| Expected result  | The case is saved and In Progress, and a unit is assigned despite notification failure.                          |
| Observed result  | Case creation and assignment continued despite the error.                                                        |
| Execution status | Pass                                                                                                             |
| Trace to C#      | Test/PrototypeTesting.cs<br>PrototypeTesting.Appendix02_NotificationFailure_DoesNotBreakCriticalDispatchWorkflow |